

A Course in Modern Macroeconomics

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Complete Solutions

Chapters 10–11 — Money and Inflation

All fourteen end-of-chapter problems, 10.1–10.5 and 11.1–11.9 (pp. 202–222)

Every problem carries the full question text, a step-by-step solution with no skipped algebra, an economic reading of the result, and a verification. Numerical answers are reproduced by the companion scripts in `Resolucao/kurlat_ch10-11_codigo/`, which check each analytical formula against an independent computation (simulated deposit chains, direct minimisation, finite differences, backward-solved price paths, root finding) and abort on any mismatch.

Problems are grouped by **topic**, not by number, following the order of the chapters' own sections. Blue boxes are odd-numbered problems, purple boxes even.

Lista 6 of the course is built on these two chapters; where a result overlaps (Baumol–Tobin demand, $\eta = \frac{1}{2}$, $\pi = \mu - \eta g$) it is derived the same way and gives the same numbers as `lista6_resolucao.tex`.

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Chapter 10 Money

Supply. The Central Bank controls the *monetary base* (currency plus reserves) directly. M1 (currency plus checking deposits) is created jointly with the banking system. If banks hold exactly the required fraction ρ of deposits as reserves and the public holds a fraction χ of its M1 as cash, one unit of new base becomes

$$\omega \equiv \frac{\Delta \text{M1}}{\Delta \text{Base}} = \frac{1}{\rho + \chi - \rho\chi} \quad (10.3.1)$$

units of M1. The formula relies on banks lending out every excess reserve, which is true only when reserves pay less than loans.

Demand (Baumol–Tobin). A household spends pY evenly over the period and pays a fixed real cost F per trip to the bank. With N trips its average balance is $M = pY/2N$ (10.4.1); minimising $pFN + i pY/2N$ (10.4.2) gives

$$N = \sqrt{\frac{iY}{2F}} \quad (10.4.3), \quad \frac{M}{p} = \sqrt{\frac{YF}{2i}} \equiv m^D(Y, i) \quad (10.4.4).$$

Real balances rise with Y and F and fall with the nominal interest rate i , the opportunity cost of holding money.

The Money Multiplier

Problem 10.1 Central Bank Instruments

Question

Suppose the Central Bank wants to reduce the M1 money supply but does not want to change the monetary base. In what direction should it change reserve requirements?

Solution:

Step 1 — M1 as base times multiplier. By (10.3.1), with the base B held fixed,

$$\text{M1} = \omega(\rho, \chi) B, \quad \omega(\rho, \chi) = \frac{1}{\rho + \chi - \rho\chi}.$$

With B fixed, M1 moves only through ω , and of its two arguments only ρ is a policy instrument (χ is the public's choice).

Step 2 — differentiate with respect to ρ . The denominator is $D = \rho + \chi - \rho\chi$, with $\partial D/\partial \rho = 1 - \chi$. Then

$$\frac{\partial \omega}{\partial \rho} = -\frac{1}{D^2} \frac{\partial D}{\partial \rho} = -\frac{1 - \chi}{(\rho + \chi - \rho\chi)^2} = -(1 - \chi)\omega^2 < 0 \quad (0 \leq \chi < 1).$$

Step 3 — answer.

Raise the reserve requirement ρ .

$$dM1 = B \frac{\partial \omega}{\partial \rho} d\rho < 0.$$

Reading. Each round of the deposit chain now leaves a larger share ρ of every new deposit idle as reserves, so less is lent out and redeposited: the geometric series $\sum_n [(1-\rho)(1-\chi)]^n$ has a smaller ratio and a smaller sum. The base does not change; it is re-split towards reserves and away from the currency that the (shorter) chain would have generated.

Example. With $\rho = 0.10$, $\chi = 0.30$: $\omega = 1/(0.10 + 0.30 - 0.03) = 1/0.37 = 2.7027$. To cut M1 by 10% the multiplier must fall to $0.9 \times 2.7027 = 2.4324$, so $\rho + \chi - \rho\chi = 1/2.4324 = 0.41111$, i.e. $\rho(1 - \chi) = 0.41111 - 0.30 = 0.11111$ and $\rho = 0.11111/0.70 = 0.1587$. The requirement must rise from 10% to about 15.9%.

Verification. The code runs Kurlat's deposit chain round by round (cash withdrawal χ , reserves ρ , re-lending of the rest) and recovers 2.7027; the finite-difference derivative matches $-(1 - \chi)\omega^2 = -5.1132$, and a root-finder returns $\rho = 0.15873$. ✓

Where this goes wrong

The answer assumes the requirement *binds*: banks hold no excess reserves. If they already hold reserves above ρd (because $i \approx 0$, or because reserves pay interest, see Problem 10.3), raising ρ up to the ratio they already hold changes nothing.

Problem 10.2 Pickpockets

Question

Suppose there is an increase in the number of pickpockets. How would that change the fraction of their money that people want to have in cash as opposed to checking deposits? If the Central Bank keeps the monetary base constant, what will happen to the M1 money supply?

Solution:

Step 1 — the currency ratio. Pickpockets raise the expected loss from carrying cash; deposits are not exposed to them. Holding cash becomes costlier relative to holding deposits, so people shift their M1 towards checking accounts:

χ falls.

Step 2 — the multiplier. From $D = \rho + \chi - \rho\chi$, $\partial D/\partial\chi = 1 - \rho$:

$$\frac{\partial\omega}{\partial\chi} = -\frac{1 - \rho}{(\rho + \chi - \rho\chi)^2} = -(1 - \rho)\omega^2 < 0.$$

A fall in χ therefore *raises* ω , and with the base fixed,

$$\boxed{\text{M1} = \omega B \text{ rises.}}$$

Step 3 — why. A dollar of base held as currency supports exactly one dollar of M1. The same dollar deposited in a bank supports $1/\rho$ dollars of deposits, because the bank needs to keep only ρ of it as reserves and lends the rest. Moving money out of pockets and into banks releases base into the reserve-lending chain.

Step 4 — the composition. The end-of-chain balance sheets give currency = $\chi\omega B$, deposits = $(1 - \chi)\omega B$ and reserves = $\rho(1 - \chi)\omega B$. With $\rho = 0.10$ and $B = 100$:

	ω	M1	Currency	Deposits	Reserves
$\chi = 0.30$	$1/0.37 = 2.7027$	270.27	81.08	189.19	18.92
$\chi = 0.20$	$1/0.28 = 3.5714$	357.14	71.43	285.71	28.57

In both rows currency + reserves = $100 = B$ (81.08 + 18.92; 71.43 + 28.57). M1 rises by 32%: currency falls, deposits rise by much more.

Verification. The simulated deposit chain reproduces every entry of the table; $\partial\omega/\partial\chi$ by finite differences equals $-(1 - \rho)\omega^2 = -6.5741$. ✓

Problem 10.3 Interest on Reserves

Question

Consider an economy where households almost use no cash (so $\chi \rightarrow 0$). Furthermore, assume that the Central Bank lowers the reserve requirement to almost zero (so $\rho \rightarrow 0$).

- What is the value of the money multiplier as χ and ρ become small?
- Suppose that the Central Bank decides to pay interest on reserves at the market rate i . Does equation (10.3.1) still hold? Why or why not?
- Under this policy regime, what determines the quantity of money?

Solution:

- The denominator of (10.3.1) goes to zero:

$$\rho + \chi - \rho\chi \rightarrow 0 + 0 - 0 = 0 \quad \Rightarrow \quad \boxed{\omega = \frac{1}{\rho + \chi - \rho\chi} \rightarrow \infty.}$$

Along $\rho = \chi = \varepsilon$, $\omega = 1/(2\varepsilon - \varepsilon^2) \approx 1/(2\varepsilon)$: 5.26 at $\varepsilon = 0.1$, 50.25 at 0.01, 500.25 at 0.001. With no cash leaking out of the banking system and no reserves required, each round of lending is redeposited in full and none of it is sterilised:

the ratio of the geometric series $(1 - \rho)(1 - \chi)$ tends to one and the sum diverges. A tiny base could in principle support an arbitrarily large M1.

- (b) **No.** Equation (10.3.1) was derived from the assumption that banks hold *exactly* the required reserves, $R = \rho d$, and lend out any excess. The reason they do so is that reserves pay zero while loans and bonds pay $i > 0$ (Kurlat’s footnote 6). Paying interest on reserves at the market rate removes that opportunity cost: a dollar of reserves now earns the same i as a dollar lent out. The bank is indifferent, and there is no longer any pressure to lend out “excess” reserves. Write $\theta \equiv R/d$ for the reserve ratio the banks actually choose. The accounting still gives

$$\frac{M1}{B} = \frac{1}{\theta + \chi - \theta\chi},$$

but now $\theta \geq \rho$ is an *endogenous and indeterminate* choice of the banks, not the policy parameter ρ . A new unit of base can simply sit on bank balance sheets as excess reserves, with $\Delta M1 = 0$. The multiplier stops being a stable number that links the base to M1. This is what Figure 10.3.1 shows after 2008: the base rose about fivefold and the M1 multiplier fell from about 2 to below 1.

- (c) The link that ran *base* $\rightarrow$ *M1* is cut, so the quantity of money is determined by **demand**. The Central Bank’s real instrument is now the interest rate it pays (equal to the market rate i). Given i , the public holds

$$M1 = p \cdot m^D(Y, i)$$

and the banks supply whatever deposits are demanded, since funding a loan or holding a reserve earns the same. The base adjusts *passively*: banks hold whatever reserves come out of this, and they hold them willingly. With $\chi \rightarrow 0$, M1 is essentially all deposits, and its size is set by the money-demand function at the interest rate the Central Bank chooses.

Where this goes wrong

“ $\omega \rightarrow \infty$, so the Central Bank has infinite leverage over M1” is the wrong conclusion to draw from (a). It holds only while banks are pushed to lend out every excess reserve. Part (b) removes exactly that push, which is why the two parts are asked together.

The Demand for Money

Question

Suppose one day, suddenly and unexpectedly, ATMs are invented, which make getting cash more convenient than before. What would be the effect of this invention on the demand for money?

Solution:

Step 1 — what changes in the model. In Baumol–Tobin “going to the bank” costs F per trip. ATMs lower it: $F \downarrow$, with Y and i unchanged.

Step 2 — trips and balances. From (10.4.3) and (10.4.4),

$$N = \sqrt{\frac{iY}{2F}} \uparrow, \quad \frac{M}{p} = \sqrt{\frac{YF}{2i}} \downarrow.$$

The elasticity of real balances with respect to F is

$$\frac{\partial \ln(M/p)}{\partial \ln F} = \frac{1}{2},$$

so a 1% fall in the trip cost lowers money demand by 0.5%. For example, if ATMs cut F by three quarters, M/p falls by the factor $\sqrt{1/4} = \frac{1}{2}$ and N doubles.

Step 3 — the demand curve. In the $(M/p, i)$ plane of Figure 10.4.2, a higher F shifts demand right; here the shift is to the **left**. At every interest rate people want to hold less money, because topping up is now cheap. They make more, smaller withdrawals and keep more wealth in interest-bearing assets.

Money demand falls at every (Y, i) : the $m^D(Y, i)$ schedule shifts left.

Step 4 — consequences (looking ahead to Chapter 11).

- *Velocity rises.* $V = Y/m^D = \sqrt{2iY/F}$ increases as F falls. The same stock of money now turns over more often.
- *Prices jump.* “Suddenly and unexpectedly” matters. If the Central Bank holds M^S fixed and prices are flexible, $p = M^S/m^D(Y, i)$ jumps up in proportion to the fall in money demand, a factor $\sqrt{F/F'}$. People try to get rid of balances they no longer want, and money loses value.
- *The supply side may move too.* If ATMs make cash more convenient relative to checks, χ may rise, which lowers the multiplier (Problem 10.2 in reverse). That is a change in money *supply* at a given base, not in the demand for M1.

Verification. The code minimises $FN + iY/2N$ numerically at F and $F/4$ and finds that N doubles and M/p halves. ✓

Problem 10.5 Going to the Bank

Question

The average person in the US has about \$10,000 in M1 money and spends around \$60,000 per year. Suppose this person uses its checking account to pay for all its purchases, behaves according to the Baumol-Tobin model and only has two investment options: its checking account (which pays zero interest) and an investment account (which pays interest). Define “going to the bank” as transferring a balance from the investment account to the checking account.

- (a) How often does the average person go to the bank?
- (b) Suppose the interest rate is 2%. What must be the perceived cost of each visit to the bank for this type of behavior to be optimal?
- (c) How often would the average person go to the bank if the interest rate rose to 3%?
- (d) Now suppose the checking account does pay interest, but it only pays 0.5% instead of the full 2% one can get from the investment account. What is the opportunity cost of holding balances in the checking account? What do we infer now about the perceived cost of going to the bank?

Solution: Work in dollars: $pY = 60,000$ per year, average balance $M = 10,000$; F below is the dollar cost of a trip (pF in Kurlat’s notation).

- (a) Invert the sawtooth average (10.4.1):

$$M = \frac{pY}{2N} \implies N = \frac{pY}{2M} = \frac{60,000}{2 \times 10,000} = \frac{60,000}{20,000} = \boxed{3 \text{ trips per year}},$$

one every four months. Each transfer is $pY/N = \$20,000$, run down to zero over four months, so the average balance is \$10,000.

- (b) The observed N must satisfy the first-order condition (10.4.3), $N^2 = iY/2F$, so

$$F = \frac{i pY}{2N^2} = \frac{0.02 \times 60,000}{2 \times 3^2} = \frac{1,200}{18} = \boxed{\$66.67 \text{ per trip}}.$$

Check. At the optimum the two cost components are equal: trips cost $FN = 66.67 \times 3 = \$200$, and forgone interest is $iM = 0.02 \times 10,000 = \200 . ✓

- (c) Keep $F = \$66.67$ and set $i = 0.03$:

$$N = \sqrt{\frac{i pY}{2F}} = \sqrt{\frac{0.03 \times 60,000}{2 \times 66.67}} = \sqrt{\frac{1,800}{133.33}} = \sqrt{13.5} = \boxed{3.67 \text{ trips per year}},$$

about one every 99 days. Equivalently, $N \propto \sqrt{i}$, so $N = 3\sqrt{3/2} = 3 \times 1.2247 = 3.674$. The average balance falls to $60,000/(2 \times 3.674) = \$8,165$.

- (d) The relevant opportunity cost is the *difference* in returns, because a dollar kept in checking still earns 0.5%:

$$i - i_{\text{checking}} = 2\% - 0.5\% = 1.5\%.$$

Re-running (b) with 0.015 in place of 0.02:

$$F = \frac{0.015 \times 60,000}{2 \times 3^2} = \frac{900}{18} = \boxed{\$50 \text{ per trip}}.$$

Check. $FN = 50 \times 3 = \$150 = 0.015 \times 10,000$. ✓ The same observed behaviour now reveals a lower trip cost. When holding money is cheaper, it takes a smaller F to justify going to the bank only three times a year.

Numerical exercise — see the companion code

The code recovers every number by *numerical* minimisation of $FN + i pY/2N$ and a root-finder on $N^*(F) = 3$: $F = 66.6667$ at $i = 2\%$; $N = 3.674235$ at $i = 3\%$; $F = 50.0000$ at an opportunity cost of 1.5%.

Chapter 11 The Price Level and Inflation

Equilibrium. Money supply equals nominal money demand, $M^S = m^D(Y, i) \cdot p$ (11.2.1), with the Fisher equation $1 + r_{t+1} = (1 + i_{t+1}) / (1 + \pi_{t+1})$, $i \approx r + \pi$ (11.1.1). In the classical view Y and r are set by real factors, so (11.2.1) determines p .

Steady states. Constant money: $p = M^S / m^D(Y, r)$ (11.2.2). Money growing at μ and Y at g , with η the income elasticity of m^D :

$$\pi = \mu - \eta g. \quad (11.2.3)$$

A one-off rise in M^S raises p in proportion. A rise in the *growth rate* raises i , lowers real balances, and so makes p *jump* on top of the faster trend.

Velocity. $V = pY/M = Y/m^D(Y, i)$ (11.2.5): any money-demand function is a theory of velocity.

Seignorage. The government budget (11.3.1) counts $M_{t+1}^B - M_t^B$ as revenue: base money is debt that pays no interest. Revenue = rate $\times$ base, and the base shrinks as inflation rises, so revenue is capped (Problem 11.6). **Costs.** Shoe-leather $\sqrt{(r + \pi)YF/2}$ (11.4.1), minimised by the Friedman rule $i = 0$, $\pi = -r$.

Money Demand and Velocity

Problem 11.1 The Elasticity of Money Demand

Question

Recall the Baumol-Tobin model of Chapter 10. Compute η , the elasticity of money demand with respect to GDP, in the money-demand function that arises from this model.

Solution:

Step 1 — the function. $m^D(Y, i) = \left(\frac{YF}{2i}\right)^{1/2}$.

Step 2 — the derivative. By the chain rule,

$$\frac{\partial m^D}{\partial Y} = \frac{1}{2} \left(\frac{YF}{2i}\right)^{-1/2} \cdot \frac{F}{2i} = \frac{1}{2} \left(\frac{YF}{2i}\right)^{1/2} \cdot \frac{1}{Y} = \frac{m^D}{2Y}.$$

Step 3 — the elasticity.

$$\eta \equiv \frac{\partial m^D}{\partial Y} \cdot \frac{Y}{m^D} = \frac{m^D}{2Y} \cdot \frac{Y}{m^D} = \boxed{\frac{1}{2}}.$$

The same follows from logs: $\ln m^D = \frac{1}{2} \ln Y + \frac{1}{2} \ln F - \frac{1}{2} \ln 2i$, so $\partial \ln m^D / \partial \ln Y = \frac{1}{2}$.

Reading. Cash management has **economies of scale**. Doubling spending raises money holdings by only $\sqrt{2}$, a 41% rise: a richer household makes more trips instead of scaling up every withdrawal. Two consequences. By (11.2.3) a growing economy absorbs money growing at $g/2$ at zero inflation, not g . And velocity $V = \sqrt{2iY/F}$ trends up with income. This is the $\eta = \frac{1}{2}$ of Lista 6, Question 2.

Verification. A central finite difference of $\ln m^D$ in $\ln Y$ returns 0.500000. ✓

Problem 11.2 The Quantity Theory

Question

The quantity theory of money assumes that money velocity is constant. Try to come up with a money-demand function that would imply that velocity is indeed constant. What does this money-demand function say about how money holdings depend on interest rates? What does it say about how households manage their trips to the bank?

Solution:

Step 1 — the function. From (11.2.5), $V = Y/m^D(Y, i)$. Velocity is a constant $\bar{V}$ for

all (Y, i) if and only if

$$m^D(Y, i) = kY, \quad k = \frac{1}{\bar{V}} > 0.$$

Check. $V = Y/(kY) = 1/k$, independent of Y and i . ✓ (This is the Cambridge form. It has $\eta = 1$, where Baumol–Tobin has $\eta = \frac{1}{2}$.)

Step 2 — interest rates. $\partial m^D / \partial i = 0$. Money holdings do **not** respond to the interest rate at all: the interest elasticity of money demand is zero. However large the opportunity cost of holding money, people hold the same balances per unit of spending. In Chapter 11's terms, a faster money growth rate would then leave real balances unchanged, and there would be no price-level jump when money growth changes and no Laffer curve for seignorage.

Step 3 — trips to the bank. In the sawtooth model average real balances are $Y/2N$. Setting $Y/2N = kY$:

$$N = \frac{1}{2k} = \frac{\bar{V}}{2} \quad (\text{a constant}).$$

Households go to the bank a **fixed number of times per period**, whatever the interest rate and however much they spend. More spending is handled by larger withdrawals, never by more trips. That is behaviour dictated by an institutional rhythm (for example, cash drawn once per pay period), not the outcome of the Baumol–Tobin trade-off.

Within Baumol–Tobin it would require $N = \sqrt{iY/2F}$ to be constant, i.e. a trip cost $F = iY/2\bar{N}^2$ that moves exactly in proportion to iY , which has no economic justification.

Reading. Constant velocity is a restriction on behaviour, not an accounting fact: it requires households to ignore the cost of holding money. The Baumol–Tobin household does not ignore it, which is why velocity in Figure 11.2.3 moves with interest rates.

Inflation in Equilibrium

Problem 11.4 Growth in the Money Supply

Question

There are two otherwise identical economies. In both of them GDP is constant, prices are flexible and the real interest rate is constant. The rate of growth of the money supply in each of them is:

Year	Country A	Country B
2009	3%	6%
2010	3%	5%
2011	3%	4%
2012	3%	3%
2013	3%	2%
2014	3%	1%
2015	3%	0%

How does inflation in the two countries in 2012 compare? Explain.

Solution: *Reading of the statement.* Both paths are known in advance (perfect foresight), and after 2015 each country keeps its last growth rate (3% in A, 0% in B).

Step 1 — equilibrium in two consecutive years. Kurlat's timing is $M_t^S = p_t m^D(Y, i_{t+1})$ with $i_{t+1} = r + \pi_{t+1}$: the balances held in t depend on the cost of holding them into $t + 1$. Dividing year t by year $t - 1$, with $1 + \mu_t = M_t^S/M_{t-1}^S$:

$$\frac{1 + \mu_t}{1 + \pi_t} = \frac{m^D(Y, r + \pi_{t+1})}{m^D(Y, r + \pi_t)} \implies 1 + \pi_t = (1 + \mu_t) \frac{m^D(Y, r + \pi_t)}{m^D(Y, r + \pi_{t+1})}.$$

Inflation equals money growth *minus the growth of real balances*, and real balances grow when expected inflation falls.

Step 2 — Country A. $\mu_t = 3\%$ forever. Guess $\pi_t = 3\%$ for all t . Then the ratio of money demands is 1, and the equation gives $1 + \pi_t = 1.03$. ✓ So $\pi_{2012}^A = \boxed{3\%}$.

Step 3 — Country B, solved backwards. From 2016 on, money is constant, so $\pi = 0$ and $i = r$.

- **2015** ($\mu = 0$, $\pi_{2016} = 0$): $1 + \pi_{2015} = m^D(r + \pi_{2015})/m^D(r)$. If $\pi_{2015} > 0$ the right side is below one, and if $\pi_{2015} < 0$ it is above one. Only $\pi_{2015} = 0$ works.
- **2014** ($\mu = 1\%$, $\pi_{2015} = 0$): $1 + \pi_{2014} = 1.01 m^D(r + \pi_{2014})/m^D(r)$. Any $\pi_{2014} > 0$ makes the ratio less than one, so $0 < \pi_{2014} < 1\%$.
- *Each earlier year* inherits a lower expected inflation for next year than this year's, $\pi_{t+1} < \pi_t$. Hence $i_{t+1} < i_t$, $m^D(r + \pi_{t+1}) > m^D(r + \pi_t)$, and the ratio in Step 1 is below one:

$$\pi_t^B < \mu_t^B \quad \text{for every year 2009–2014.}$$

In 2012, $\mu^B = 3\%$, so

$$\boxed{\pi_{2012}^B < 3\% = \pi_{2012}^A.}$$

Reading. Money growth is the same in 2012, but the future is not. People in B know money growth, and therefore inflation, will keep falling, so the nominal interest rate is

falling and they want to hold *rising* real balances. Part of B's new money is absorbed by that rising demand instead of pushing prices up. The price level is forward-looking: it depends on the whole future path of money, not on this year's growth rate alone. This is the “change in the rate of growth” logic of §11.2 applied gradually.

Numerical exercise — see the companion code

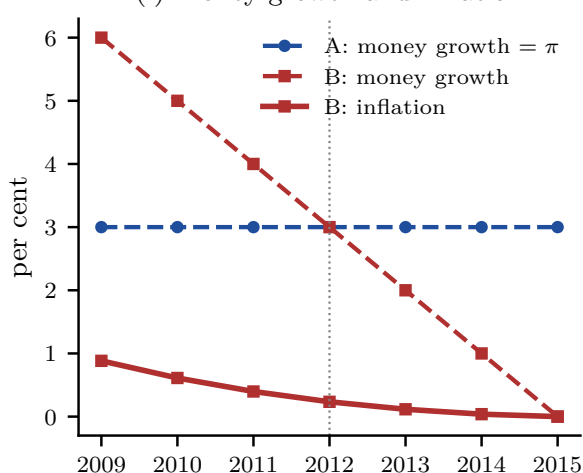
With Baumol–Tobin demand and $r = 2\%$, the code solves p_t backwards year by year by root finding and checks that the money market clears in every year:

	2009	2010	2011	2012	2013	2014	2015
μ^B	6%	5%	4%	3%	2%	1%	0%
π^B	0.88%	0.61%	0.40%	0.23%	0.12%	0.04%	0.00%

$\pi^A = 3.000\%$ in every year. The *size* of the gap depends on how interest-sensitive money demand is; the *sign* does not.

11.4 — Baumol–Tobin demand, $r = 2\%$, path fully anticipated

(i) Money growth and inflation



(ii) Real balances in B rise

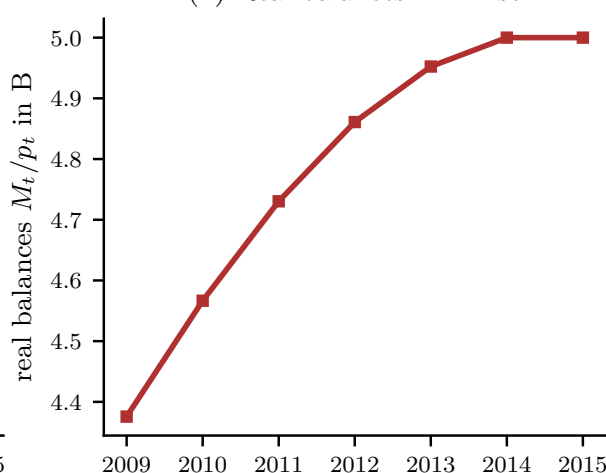


Figure 1: Problem 11.4. Money growth and inflation in A and B (left); B's real balances (right).

Where this goes wrong

“Both grow money at 3% in 2012, so both have 3% inflation” applies the steady-state formula $\pi = \mu$ out of steady state. $\pi = \mu$ holds only when expected inflation, and hence i and real balances, are constant. Country A is in that position; Country B is not.

Problem 11.5 Inflation Targeting

Question

- (a) Suppose a Central Bank has decided it wants to keep inflation at exactly 2% every year. The economy is not growing and the real interest rate is 3%. Describe what the Central Bank must do to the money supply to achieve its objective.
- (b) Suppose the real interest rate suddenly and unexpectedly falls from 3% to 1%. How should the Central Bank respond? Explain in detail.

Solution:

- (a) With $g = 0$, (11.2.3) gives $\pi = \mu$. The Central Bank must

$$\text{grow the money supply at } \mu = 2\% \text{ per year: } M_{t+1}^S = 1.02 M_t^S.$$

Verification. The nominal rate is $i = r + \pi = 3\% + 2\% = 5\%$ (exactly $1.03 \times 1.02 - 1 = 5.06\%$), constant. Real balances $m^D(Y, 0.05)$ are constant, so $p_t = M_t^S / m^D(Y, 0.05)$ grows exactly as M_t^S does, at 2%. ✓ The *level* of M^S pins the level of p ; its *growth rate* pins inflation.

- (b) **Step 1 — what the shock does to money demand.** If inflation stays at 2%, the nominal rate falls from $3\% + 2\% = 5\%$ to $1\% + 2\% = 3\%$. Holding money is cheaper, so real money demand rises from $m^D(Y, 0.05)$ to $m^D(Y, 0.03)$.

Step 2 — what happens if the Central Bank does nothing. The stock $M_t^S = 1.02 M_{t-1}^S$ is fixed at the moment of the shock. Then

$$\frac{p_t}{p_{t-1}} = \frac{M_t^S / m^D(Y, 0.03)}{M_{t-1}^S / m^D(Y, 0.05)} = 1.02 \times \frac{m^D(Y, 0.05)}{m^D(Y, 0.03)} < 1.02.$$

With Baumol–Tobin demand, $m^D \propto i^{-1/2}$, so the ratio is $\sqrt{0.03/0.05} = 0.7746$ and $p_t/p_{t-1} = 1.02 \times 0.7746 = 0.790$: a one-off **deflation of about 21%**. People try to build up balances, nobody can, and the value of money rises. The target is missed badly.

Step 3 — the right response: a one-time level increase, then the old growth rate. Choose M_t^S so that $p_t = 1.02 p_{t-1}$:

$$M_t^S = 1.02 p_{t-1} m^D(Y, 0.03) = 1.02 M_{t-1}^S \frac{m^D(Y, 0.03)}{m^D(Y, 0.05)},$$

and from then on $M_{t+s}^S = 1.02 M_{t+s-1}^S$ again. With Baumol–Tobin,

$$\frac{m^D(Y, 0.03)}{m^D(Y, 0.05)} = \sqrt{\frac{0.05}{0.03}} = \sqrt{\frac{5}{3}} = 1.2910,$$

a one-time injection of 29.1% on top of the 2% trend, at the moment of the shock; then 2% growth as before.

Step 4 — why this is the whole answer. After the shock the economy is again in a steady state with constant $i = 3\%$, so real balances are constant and 2% money growth gives 2% inflation. Only the *level* of money demand has changed, and a level change in demand is met by a level change in supply. Keeping 2% growth without the jump misses the target for one year (Step 2). Raising the growth rate instead would raise inflation permanently. Timing matters: the injection must come with the shock, because a delay means a fall in prices followed by a jump back.

Verification. The code solves the perfect-foresight price path after the shock for a trial jump x , finds by root-finding the x that gives $\pi = 2\%$ in the shock year, $x = 1.290994 = \sqrt{5/3}$, confirms $\pi = 2\%$ in every later year, and finds $\pi = -20.99\%$ with no jump. ✓

Seignorage

Problem 11.3 Seignorage with Zero Inflation and Growth

Question

Suppose an economy is growing at a constant rate g , money demand comes from the Baumol-Tobin model and the money multiplier is ω . Suppose the government wants to maintain zero inflation. How much seignorage revenue can it obtain as a fraction of GDP? In other words, find an expression for the level of:

$$\frac{M_t^B - M_{t-1}^B}{p_t Y_t}$$

that is consistent with zero inflation. How does this depend on g , F and ω ? Why? [Hint: use the result from Exercise 11.1.]

Solution:

Step 1 — what zero inflation pins down. $\pi = 0$ means $p_t = p$ constant and, by the Fisher equation, $i = r$ constant. Money-market equilibrium in every period:

$$M_t^S = p m^D(Y_t, r) = p \sqrt{\frac{Y_t F}{2r}}.$$

The Central Bank must supply exactly this; the base is $M_t^B = M_t^S / \omega$.

Step 2 — growth of the base. With $Y_{t-1} = Y_t / (1 + g)$,

$$M_{t-1}^B = \frac{p}{\omega} \sqrt{\frac{Y_{t-1} F}{2r}} = \frac{p}{\omega} \sqrt{\frac{Y_t F}{2r}} (1 + g)^{-1/2} = M_t^B (1 + g)^{-1/2}.$$

The base grows at $(1 + g)^{1/2} - 1 \approx g/2 = \eta g$: this is (11.2.3) with $\pi = 0$, $\mu = \eta g$, and $\eta = \frac{1}{2}$ from Problem 11.1.

Step 3 — revenue over GDP.

$$\frac{M_t^B - M_{t-1}^B}{pY_t} = \frac{M_t^B [1 - (1 + g)^{-1/2}]}{pY_t} = \frac{1}{\omega} \cdot \frac{1}{Y_t} \sqrt{\frac{Y_t F}{2r}} [1 - (1 + g)^{-1/2}],$$

$$\boxed{\frac{M_t^B - M_{t-1}^B}{p_t Y_t} = \frac{1}{\omega} \sqrt{\frac{F}{2r Y_t}} [1 - (1 + g)^{-1/2}] \approx \frac{\eta g}{\omega} \cdot \frac{m^D(Y_t, r)}{Y_t} = \frac{g}{2\omega} \sqrt{\frac{F}{2r Y_t}}.}$$

The approximation uses $1 - (1 + g)^{-1/2} \approx g/2$ for small g .

Step 4 — comparative statics and why.

- **Increasing in g .** Growth raises the demand for real balances every period, and the government meets it by issuing new base *without* causing inflation. With no growth ($g = 0$) the revenue is zero: every dollar of seignorage then requires inflation (footnote 4 of the chapter). The hint enters here: revenue is proportional to ηg ; with $\eta = 1$ it would be twice as large.
- **Increasing in F** (as $\sqrt{F}$). A higher trip cost makes people hold more money per unit of GDP, $m^D/Y = \sqrt{F/2rY}$. The tax base, and its growth, are larger. Financial innovation that lowers F erodes the revenue.
- **Decreasing in ω** (as $1/\omega$). The government issues only the base. The rest of M1 is deposits, which private banks create, and the interest saved on them accrues to banks (Problem 11.7). The larger the multiplier, the smaller the base behind each dollar of M1.

One more feature: because $\eta < 1$, $m^D/Y \propto Y^{-1/2}$ falls as the economy grows, so the revenue *share* slowly declines over time even at constant g .

Numerical exercise — see the companion code

With $g = 0.03$, $F = 0.02$, $\omega = 5$, $r = 0.03$, the code builds Y_t , M_t^S and M_t^B period by period at $p = 1$ and matches the exact formula to 10^{-14} : 0.0015734 of GDP, against 0.0016087 from the first-order approximation (a gap of 2.2%, of order g). It also confirms that revenue rises with g and F (a fourfold F doubles it), falls with ω , and is zero at $g = 0$.

Problem 11.6 Seignorage with High Inflation

Question

Suppose that money demand is given by:

$$m^D(Y_t, i_{t+1}) = Y_t \cdot (1 + i_{t+1})^{-\mu}$$

where μ is a parameter. This is sometimes known as the Cagan money-demand function. Equilibrium in the money market is given by:

$$M_t^S = Y_t \cdot (1 + i_{t+1})^{-\mu} p_t$$

The real economy is in a steady state so that $Y_t = Y_{ss}$, $r_{t+1} = r_{ss}$. The government obtains seignorage revenue S_t (in real terms) by expanding the monetary base. S_t is given by:

$$S_t = \frac{M_t^B - M_{t-1}^B}{P_t} \quad (11.4.2)$$

Let ω be the money multiplier, so that $M^S = \omega M^B$. Suppose that the rate of growth of the money supply is constant at rate γ , i.e.:

$$M_t^S = (1 + \gamma) M_{t-1}^S \quad (11.4.3)$$

- (a) What will be the rate of inflation?
- (b) What will be nominal interest rate?
- (c) Find an expression for the level of real money balances. How does it depend on γ ? Why?
- (d) Solve equation (11.4.3) for M_{t-1}^S and replace this in (11.4.2) to obtain an expression for seignorage revenue S_t in terms of γ and $\frac{M_t^S}{P_t}$. How does S_t depend on γ ? How does it depend on $\frac{M_t^S}{P_t}$? Why?
- (e) Replace the value of $\frac{M_t^S}{P_t}$ that you found in part (c) into the expression for S_t that you found in part (d) to obtain an expression for seignorage revenues S_t in terms of γ , Y_{ss} , r_{ss} , ω and μ .
- (f) Assume the following parameter values: $Y_{ss} = 1$, $\mu = 4$, $r_{ss} = 0.0025$, $\omega = 5$. Note that $\mu = 4$ is close to the value that has been empirically estimated in some high inflation countries when the time period is one month. To be consistent, r_{ss} should be interpreted as a monthly real interest rate and Y_{ss} as monthly GDP.
- (g) Plot S_t against γ for values of γ between 0 and 0.8.
- (h) What is the monthly rate of growth of the money supply that maximizes seignorage revenue? Call this γ^* .

Question (continued)

- (i) Why is seignorage revenue decreasing in γ for $\gamma > \gamma^*$?
- (j) What monthly inflation rate does $\gamma = \gamma^*$ imply?
- (k) What monthly nominal interest rate does $\gamma = \gamma^*$ imply?
- (l) What yearly rate of inflation does $\gamma = \gamma^*$ imply?
- (m) What is $\frac{M_t}{P_t}$ if $\gamma = \gamma^*$?
- (n) What is the maximum revenue as a fraction of GDP that the government in this example can constantly collect from seignorage?
- (o) Look at the data from Sargent (1982) on the German hyperinflation of the 1920s (you can find it at the book website). What was the maximum monthly inflation rate that Germany experienced? How does it compare to γ^* ?
- (p) Suppose that the government is increasing the money supply at a rate $\gamma = \gamma^*$, and in month t it credibly announces that (i) M_{t+1}^B will not be equal to $M_t^B(1 + \gamma^*)$ the way it would have been if the policy had continued as usual, but it will be some other level $\bar{M}$ (maybe higher, maybe lower than $M_t^B(1 + \gamma^*)$) (ii) from then on, it will set $M_{t+s}^B = \bar{M}$ for every $s \geq 1$ (i.e. the money supply will be constant).
- (q) What will be the rate of inflation from period $t + 1$ onwards?
- (r) What will be the price level P_{t+1} in period $t + 1$? (This will depend on the government's choice of $\bar{M}$.)
Suppose the government sets $\bar{M}$ at the level that will ensure that $P_{t+1} = P_t$ (i.e. the level that will immediately stop inflation).
- (s) What value of $\bar{M}$ will it need to choose? How does it compare to $M_t^B(1 + \gamma^*)$?
- (t) How much seignorage revenue will the government obtain in period $t + 1$? How does this level of seignorage compare to what the government was getting every month before making this change?
- (u) Look carefully at the timing of the end of hyperinflation in Germany. When exactly did the money supply stop growing? When exactly did hyperinflation stop? How can we make sense of this?

Solution: In this problem μ is Cagan's parameter, the elasticity of real balances with respect to $1 + i$, not a money growth rate. Money growth is γ .

- (a) **Guess and verify.** Guess a steady state with constant inflation π . Then i_{t+1} is constant, so real balances $M_t^S/P_t = Y_{ss}(1 + i)^{-\mu}$ are constant, and P_t must grow at the same rate as M_t^S :

$$\frac{P_t}{P_{t-1}} = \frac{M_t^S}{M_{t-1}^S} = 1 + \gamma \quad \implies \quad \boxed{\pi = \gamma.}$$

This is (11.2.3) with $g = 0$.

- (b) Fisher, exactly: $1 + i = (1 + r_{ss})(1 + \pi)$, so

$$\boxed{i = (1 + r_{ss})(1 + \gamma) - 1 \approx r_{ss} + \gamma.}$$

- (c) From the equilibrium condition with $1 + i = (1 + r_{ss})(1 + \gamma)$:

$$\boxed{\frac{M_t^S}{P_t} = Y_{ss} [(1 + r_{ss})(1 + \gamma)]^{-\mu}}, \quad \frac{\partial \ln(M^S/P)}{\partial \ln(1 + \gamma)} = -\mu < 0.$$

Real balances **fall** with γ . Faster money growth means higher inflation, a higher nominal rate, a higher opportunity cost of holding money, and so smaller real holdings.

- (d) From (11.4.3), $M_{t-1}^S = M_t^S / (1 + \gamma)$; with $M^B = M^S / \omega$,

$$S_t = \frac{M_t^S - M_{t-1}^S}{\omega P_t} = \frac{1}{\omega} \left(1 - \frac{1}{1 + \gamma} \right) \frac{M_t^S}{P_t} \Rightarrow \boxed{S_t = \frac{1}{\omega} \cdot \frac{\gamma}{1 + \gamma} \cdot \frac{M_t^S}{P_t}}.$$

Holding real balances fixed, S_t is **increasing in** γ : $\gamma/(1 + \gamma)$ is the share of this period's stock that is newly printed, the *rate* of the inflation tax. It is **proportional to** M_t^S/P_t , the *base* of the tax: a given growth rate collects more when people hold more money. Revenue = rate $\times$ base $\div \omega$.

- (e) Substitute (c) into (d):

$$\boxed{S = \frac{Y_{ss}}{\omega} \cdot \frac{\gamma}{1 + \gamma} [(1 + r_{ss})(1 + \gamma)]^{-\mu} = \frac{Y_{ss}}{\omega} (1 + r_{ss})^{-\mu} \gamma (1 + \gamma)^{-(1+\mu)}}.$$

Now γ raises the rate but shrinks the base.

- (f) With $Y_{ss} = 1$, $\mu = 4$, $r_{ss} = 0.0025$, $\omega = 5$: $(1.0025)^{-4} = 0.990062$ and

$$S(\gamma) = \frac{0.990062}{5} \gamma (1 + \gamma)^{-5} = 0.198012 \gamma (1 + \gamma)^{-5}.$$

- (g) See Figure 2: an inflation-tax Laffer curve, zero at $\gamma = 0$, a single peak, then slowly declining.

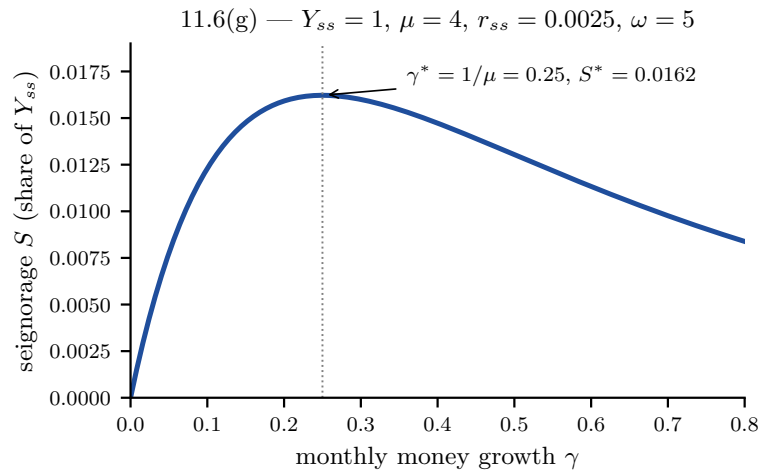


Figure 2: Problem 11.6(g). Seignorage against monthly money growth.

- (h) Maximise $h(\gamma) = \gamma(1 + \gamma)^{-(1+\mu)}$ (the constant factor does not move the argmax):

$$\begin{aligned} h'(\gamma) &= (1 + \gamma)^{-(1+\mu)} - (1 + \mu)\gamma(1 + \gamma)^{-(2+\mu)} \\ &= (1 + \gamma)^{-(2+\mu)} [(1 + \gamma) - (1 + \mu)\gamma] = (1 + \gamma)^{-(2+\mu)} (1 - \mu\gamma). \end{aligned}$$

Setting it to zero:

$$\gamma^* = \frac{1}{\mu} = \frac{1}{4} = 25\% \text{ per month.}$$

$h' > 0$ for $\gamma < 1/\mu$ and $h' < 0$ for $\gamma > 1/\mu$, so this is the global maximum.

- (i) Take logs of (e) and differentiate with respect to $\ln(1 + \gamma)$:

$$\frac{d \ln S}{d \ln(1 + \gamma)} = \underbrace{\frac{d \ln \frac{\gamma}{1+\gamma}}{d \ln(1 + \gamma)}}_{\text{rate: } 1/\gamma} + \underbrace{\frac{d \ln(M^S/P)}{d \ln(1 + \gamma)}}_{\text{base: } -\mu} = \frac{1}{\gamma} - \mu,$$

where the rate term is $(1 + \gamma) \left[\frac{1}{\gamma} - \frac{1}{1+\gamma} \right] = \frac{1}{\gamma}$. For $\gamma > 1/\mu$, $1/\gamma < \mu$: a further rise in money growth raises the tax rate by proportionally *less* than it shrinks the tax base. People flee money (more trips to the bank, foreign currency, goods) faster than the government can tax it. This is the Laffer logic the chapter describes in §11.3.

- (j) $\pi = \gamma^* = \boxed{25\% \text{ per month}}$.

- (k) $i = (1.0025)(1.25) - 1 = 1.253125 - 1 = \boxed{25.31\% \text{ per month}}$.

- (l) Compound twelve months: $1.25^2 = 1.5625$, $1.25^4 = 2.44140625$, $1.25^8 = 5.96046448$, $1.25^{12} = 5.96046448 \times 2.44140625 = 14.5519152$, so

$$\boxed{(1.25)^{12} - 1 = 13.55, \text{ i.e. about } 1,355\% \text{ per year.}}$$

- (m) $1.253125^2 = 1.57032227$, $1.253125^4 = 1.57032227^2 = 2.46591202$, so

$$\frac{M_t}{P_t} = 1 \times (1.253125)^{-4} = \frac{1}{2.46591202} = \boxed{0.4055}$$

(59% below the $(1.0025)^{-4} = 0.990$ held at zero inflation).

- (n) $S^* = \frac{1}{5} \cdot \frac{0.25}{1.25} \cdot 0.405529 = 0.2 \times 0.2 \times 0.405529 = \boxed{0.01622}$, i.e. **1.62% of GDP** (monthly revenue over monthly GDP), the most the government can collect month after month.

- (o) The Sargent (1982) tables are on the book's website and are not in this repository, so the exact monthly figures are not reproduced here. The magnitude is well documented: at the peak, in the autumn of 1923, German prices were rising about

300-fold in a single month (Jones, *Macroeconomics*, 2020, ch. 8, drawing on Sargent), i.e. monthly inflation near 29,900%, against $\gamma^* = 25\%$: **roughly 1,200 times** the revenue-maximising rate. Germany was far down the wrong side of the Laffer curve. At such rates real balances are a tiny fraction of their normal level and steady-state revenue is far below S^* . Printing that fast makes sense only as a chase of *transitory* revenue (each acceleration collects more before money demand adjusts) or as a loss of fiscal control. It cannot be explained as steady-state revenue maximisation.

(p) (Set-up of the reform; the questions are (q)–(t).)

(q) From $t + 1$ on, M^B , and so $M^S = \omega \bar{M}$, is constant. By (a) with $\gamma = 0$:

$$\pi_{t+2} = \pi_{t+3} = \dots = 0, \quad i_{t+2} = r_{ss}.$$

Inflation between t and $t + 1$ depends on $\bar{M}$ (parts (r)–(s)).

(r) Money-market equilibrium in $t + 1$, with $1 + i_{t+2} = 1 + r_{ss}$:

$$\omega \bar{M} = Y_{ss}(1 + r_{ss})^{-\mu} P_{t+1} \quad \Rightarrow \quad P_{t+1} = \frac{\omega \bar{M} (1 + r_{ss})^\mu}{Y_{ss}}.$$

(s) *Reading of the timing.* The month- t market has already cleared under the old expectations (announcement at the end of month t), so $P_t = \omega M_t^B [(1 + r_{ss})(1 + \gamma^*)]^\mu / Y_{ss}$. Setting $P_{t+1} = P_t$:

$$\begin{aligned} \omega \bar{M} (1 + r_{ss})^\mu &= \omega M_t^B (1 + r_{ss})^\mu (1 + \gamma^*)^\mu \\ \Rightarrow \quad \bar{M} &= M_t^B (1 + \gamma^*)^\mu = 1.25^4 M_t^B = 2.441 M_t^B. \end{aligned}$$

Compared with the business-as-usual $M_t^B (1 + \gamma^*) = 1.25 M_t^B$, the ratio is $(1 + \gamma^*)^{\mu-1} = 1.25^3 = 1.953$. To **stop** inflation the government must print **almost twice as much** as it would have printed to *continue* it. The reason is that ending inflation drops the nominal rate from 25.3% to 0.25% a month, so real money demand jumps by the factor $1.25^4 = 2.44$ (remonetisation). At constant prices, that extra demand can only be met with extra nominal money. If the government printed less, P_{t+1} would *fall*.

(t)

$$\begin{aligned} S_{t+1} &= \frac{\bar{M} - M_t^B}{P_{t+1}} = \frac{M_t^B}{P_t} [(1 + \gamma^*)^\mu - 1] \\ &= \frac{0.405529}{5} \times (2.44140625 - 1) = 0.081106 \times 1.44140625 = \boxed{0.1169}, \end{aligned}$$

using $P_{t+1} = P_t$ and $M_t^B / P_t = (M_t^S / P_t) / \omega$. That is **11.7% of monthly GDP**, $0.1169 / 0.01622 = \mathbf{7.21}$ times the steady-state maximum S^* (exactly $(1 + \gamma^*)[(1 +$

$\gamma^*)^\mu - 1]/\gamma^* = 1.25 \times 1.4414/0.25$). It is a one-off: from $t + 2$ on revenue is zero. A credible stabilisation hands the government a single large seignorage payment, collected from the public's rebuilding of its real balances.

11.6(p)–(t) — ending a hyperinflation at $\gamma^* = 0.25$

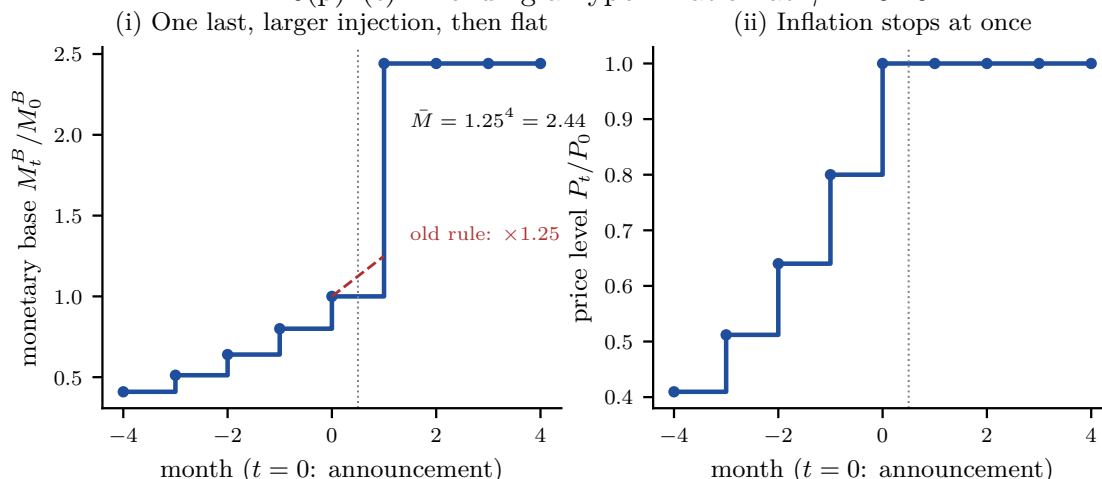


Figure 3: Problem 11.6(p)–(t). The base jumps to $\bar{M} = 2.44 M_t^B$ and stays there; the price level stops rising at once.

- (u) Sargent's reading of the German episode (and of the other interwar hyperinflations) is that inflation ended **abruptly** in November 1923, with the currency reform and a credible fiscal reform, while the stock of central-bank money **kept growing substantially for months afterwards**. Prices stopped before money did. The model explains the timing. Once the regime change is credible, expected inflation collapses, the nominal rate falls, and the demand for real balances multiplies, by 1.25^4 in our calibration and by far more from German inflation rates. Large nominal injections after stabilisation are then exactly what *keeps* prices stable: they meet the remonetisation demand of part (s). What had to stop at once was the expected *future* path of money (fiscal credibility), not the current printing.

Numerical exercise — see the companion code

The code builds M_t^S , M_t^B and P_t from the definitions and matches (e) to 10^{-14} ; a bounded optimiser and an 800,001-point grid both return $\gamma^* = 0.250000$. It prints $i = 0.253125$, $1.25^{12} - 1 = 13.5519$, $M/P = 0.405529$, $S^* = 0.016221$; the base elasticity -4 and rate elasticity $1/\gamma^* = 4$ at the peak by finite differences; and, by root-finding on $P_{t+1} = P_t$, $\bar{M} = 2.441406 M_t^B$, $S_{t+1} = 0.116907$ and the ratio 7.2070.

Where this goes wrong

The timing in (s) is a reading, not a given. If the announcement came *before* the month- t market cleared, then i_{t+1} would already reflect zero expected inflation, P_t itself would fall on the announcement (to $\omega M_t^B(1 + r_{ss})^\mu / Y_{ss}$), and $P_{t+1} = P_t$ would require $\bar{M} = M_t^B$: no extra money at all. The question's comparison with $M_t^B(1 + \gamma^*)$ and part (u) point to the reading used above, where P_t is already set under the old regime.

Also: (q) asks about inflation “from $t + 1$ onwards”. Inflation between $t + 1$ and later months is zero whatever $\bar{M}$ is; inflation *into* $t + 1$ is what $\bar{M}$ controls.

Problem 11.7 Bank Seignorage*Question*

Suppose that the demand for M1 money is given by $\frac{M}{pY} = A - \eta i$ where M is the quantity of M1 money, pY is nominal GDP, i is the nominal interest rate and A and η are parameters. Households want to hold a fraction χ of their M1 money in the form of cash and a fraction $1 - \chi$ in the form of checking accounts, which earn no interest. Banks earn seignorage by taking checking deposits and investing in assets that earn the nominal interest rate.

- Find an expression for the ratio of total seignorage earned by banks to GDP. Call this ratio s .
- Compute $\frac{\partial s}{\partial i}$. Why does this number depend on η ?
- Look up data on M1 and its components in 2018. What is a reasonable value for χ ? What was the value of $\frac{M}{pY}$? If the average nominal interest was 2%, how much seignorage did banks earn as a fraction of GDP?
- If $\eta = 0.2$, how much seignorage would banks earn as a fraction of GDP if the nominal interest rate went up to 3%?

Solution: As in the statement, banks invest the whole deposit base at i (no reserves; with a reserve ratio ρ earning nothing, multiply every expression by $1 - \rho$). Note that η here is the semi-elasticity in $A - \eta i$, not the income elasticity of §11.2.

- Deposits are $D = (1 - \chi)M$. Banks pay zero on them and earn i , so bank seignorage is $iD = i(1 - \chi)M$. Dividing by pY and substituting the demand function:

$$s = \frac{i(1 - \chi)M}{pY} = i(1 - \chi)(A - \eta i).$$

-

$$\frac{\partial s}{\partial i} = (1 - \chi)[(A - \eta i) + i(-\eta)] = (1 - \chi)(A - 2\eta i).$$

A higher i has two effects. It raises the margin on every deposit dollar (the term $A - \eta i$, the current base), and it shrinks the base, because people hold less M1

when its opportunity cost rises (the term $-\eta i$). η measures how fast the base shrinks. With $\eta = 0$, bank seignorage would be linear in i ; with $\eta > 0$ it is a Laffer curve in i with peak at $i = A/(2\eta)$, the same logic as Problem 11.6.

- (c) *Data (approximate, US, 2018, FRED series CURRSL, M1SL, GDP; pre-2020 M1 definition).* Currency $\approx$ \$1.63 trillion, M1 $\approx$ \$3.75 trillion, nominal GDP $\approx$ \$20.6 trillion. Then

$$\chi \approx \frac{1.63}{3.75} = \boxed{0.43}, \quad \frac{M}{pY} \approx \frac{3.75}{20.6} = \boxed{0.18},$$

$$s = 0.02 \times (1 - 0.435) \times 0.182 = \boxed{0.0021}, \text{ about } 0.2\% \text{ of GDP } (\approx \$42 \text{ billion}).$$

Caveats: a large share of US currency is held abroad, so χ overstates domestic cash use; and some checkable deposits do pay a little interest, which lowers the true margin.

- (d) Calibrate A from (c): $A = M/pY + \eta i = 0.1820 + 0.2 \times 0.02 = 0.1860$. At $i = 3\%$:

$$\frac{M}{pY} = 0.1860 - 0.2 \times 0.03 = 0.1800, \quad s = 0.03 \times 0.565 \times 0.1800 = \boxed{0.0031},$$

about **0.31% of GDP**, 48% more than at 2%. The base shrinks only 1.1%, and the margin rises 50%, because $i = 3\%$ is far below the peak $A/2\eta = 46.5\%$.

Verification. The code builds M , D and interest income in levels and matches (a); the finite-difference derivative matches (b); it reproduces $\chi = 0.435$, $M/pY = 0.1820$, $s(2\%) = 0.00206$, $A = 0.18604$ and $s(3\%) = 0.00305$ (ratio 1.4835). ✓

Real Interest Rates

Problem 11.8 Real Interest Rates

Question

Define realized real interest rates r_{t+1} as:

$$r_{t+1} \equiv i_{t+1} - \pi_{t+1}$$

where i_{t+1} is the nominal interest rate between period t and period $t + 1$ and π_{t+1} is the rate of inflation between period t and period $t + 1$. If there is uncertainty about what the rate of inflation is going to be, this implies that there is uncertainty about what realized real interest rates will be. Define the expected real interest rate as:

$$r_{t+1}^E \equiv i_{t+1} - E(\pi_{t+1})$$

where $E(\pi_{t+1})$ is the expected rate of inflation. Suppose there are two parties to a nominal lending contract: a borrower and a lender. Who benefits when realized real interest rates turn out to be higher than expected real interest rates? What monetary policy will each of them support?

Solution:

Step 1 — the gap. The nominal rate i_{t+1} is fixed in the contract, so subtracting the two definitions,

$$r_{t+1} - r_{t+1}^E = (i_{t+1} - \pi_{t+1}) - (i_{t+1} - E(\pi_{t+1})) = E(\pi_{t+1}) - \pi_{t+1}.$$

Realized real rates exceed expected ones exactly when **inflation turns out lower than expected**.

Step 2 — who gains. Per unit of goods lent, the lender receives a real return r_{t+1} and the borrower pays that same r_{t+1} . The contract was priced on r_{t+1}^E . When $r_{t+1} > r_{t+1}^E$ the lender gets more goods back than bargained for, and the borrower repays more goods than bargained for:

the **lender** gains and the **borrower** loses, each by $E(\pi_{t+1}) - \pi_{t+1}$ per unit lent.

Example: $i = 6\%$, $E(\pi) = 3\%$, so $r^E = 3\%$. If $\pi = 1\%$, $r = 5\%$: the lender earns 2 extra cents per real dollar and the borrower pays them. If $\pi = 5\%$, $r = 1\%$ and the transfer reverses.

Step 3 — policy preferences. Only *unexpected* inflation redistributes; expected inflation is already in i .

- **Lenders** (creditors, bondholders, savers in nominal assets) gain from inflation below expectations and lose from surprises above. They favour a tight, anti-inflationary central bank, and at the very least one that never lets inflation exceed what was priced in.
- **Borrowers** (debtors with fixed-rate nominal debt, including governments) gain from inflation above expectations, which erodes the real value of their repayments. They favour looser policy and higher-than-expected inflation.

Both sides have the same interest *ex ante* in predictable inflation, which lowers the risk on either side of the contract. The conflict arises *ex post*, once the contracts are signed.

Money in Practice

Problem 11.9 Money among Prisoners of War

Question

Read “The Economic Organisation of a P.O.W. Camp”, which you can find at: <https://www.jstor.org/stable/2550133>. Pick one passage out of the article and explain how it relates to the models of money supply, money demand and inflation from Chapters 10 and 11.

Solution: The article (Radford, *Economica*, 1945) is not in this repository, so the passage is *paraphrased* below, not quoted; the reader should check the wording against the original.

The passage. Radford describes how cigarettes became the camp’s money, and how the general level of prices, measured in cigarettes, rose and fell with the supply of cigarettes. When a large delivery of Red Cross parcels or private cigarette consignments arrived, prices rose. As the week went on and cigarettes were smoked, prices fell. Long interruptions in the supply of cigarettes produced a general deflation.

Mapping onto Chapters 10–11.

- (a) *Money supply.* The camp’s money stock M^S is the stock of cigarettes held for exchange. No authority controls it: it rises with deliveries (an “open market operation” by the Red Cross) and *shrinks through consumption*, since smoking a cigarette destroys money. Commodity money has an intrinsic use that competes with its monetary role.
- (b) *Money demand.* Real demand $m^D(Y, i)$ depends on the volume of transactions (the goods in the parcels being traded) and on the cost of holding cigarettes as money instead of smoking them. That cost plays the role of the forgone interest i in Baumol–Tobin. Within a parcel cycle the volume of goods to trade is roughly fixed, so real demand is roughly stable.
- (c) *Price level.* With stable real demand and a flexible price system, (11.2.2) gives $p = M^S/m^D(Y, i)$: the camp’s price level moves in proportion to its money stock. A delivery is a one-time increase in M^S and prices jump (§11.2, one-time increase in the money supply). Smoking during the week is a steady *negative* μ and gives steady deflation, $\pi \approx \mu$ in (11.2.3) with $g \approx 0$. This is the quantity theory working in the most transparent laboratory one could design, with flexible prices and no interest-bearing assets to muddy velocity.

- (d) *A further link.* The same article describes how cigarettes of worse quality circulated as money while better ones were smoked (Gresham's law). The *quality* of the money stock, not only its quantity, reacted to the incentive to use the good-for-smoking unit as a good, not as money.

Companion code

Every numerical result above is reproduced by `Resolucao/kurlat_ch10-11_codigo/`. Each script checks its analytical formulas against an independent numerical computation and aborts on any mismatch.

File	What it does
<code>ch10_11_numerico.py</code>	10.1–10.3: a round-by-round deposit chain against (10.3.1), the finite-difference derivatives in ρ and χ , the table of Problem 10.2 and the divergence of ω . 10.4–10.5: Baumol–Tobin by numerical minimisation and root-finding. 11.1: η by finite differences. 11.3: period-by-period base against the closed form. 11.4 and 11.5: perfect-foresight price paths solved backwards by root-finding. 11.6: seignorage built from the definitions, the argmax by optimiser and grid, all of (h)–(n), and $\bar{M}$ and S_{t+1} by root-finding. 11.7: levels against the formula, the derivative, the 2018 numbers. 11.8: the sign of the transfer.
<code>ch10_11_figuras.py</code>	The three figures.
<code>estilo_mpl.py</code>	<code>pgf</code> backend, so figure text is typeset by <code>pdflatex</code> with <code>lmodern</code> and <code>cmap</code> and the PDF stays fully copyable.